

Bird Brain

Group Members:

Aaron Duong, Koa Lister, Vinnie De Lisi, Jackson Gothie, Hunter Irish



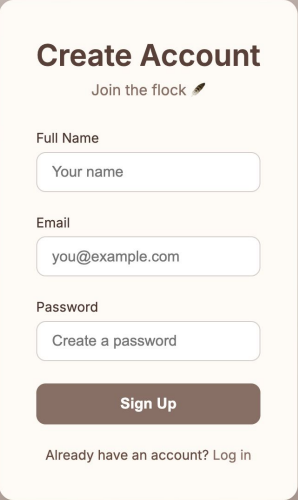
Agenda

1. Introduction
2. Tools
3. Architecture Diagram
4. Challenges
5. Future Scope/Enhancements
6. Demo

Web application with login, registration,
and personalized sessions

Designed to feel simple, familiar, and
easy to navigate

Built around one central purpose: bird
logging and sight exploration



Create Account
Join the flock 🐦

Full Name

Email

Password

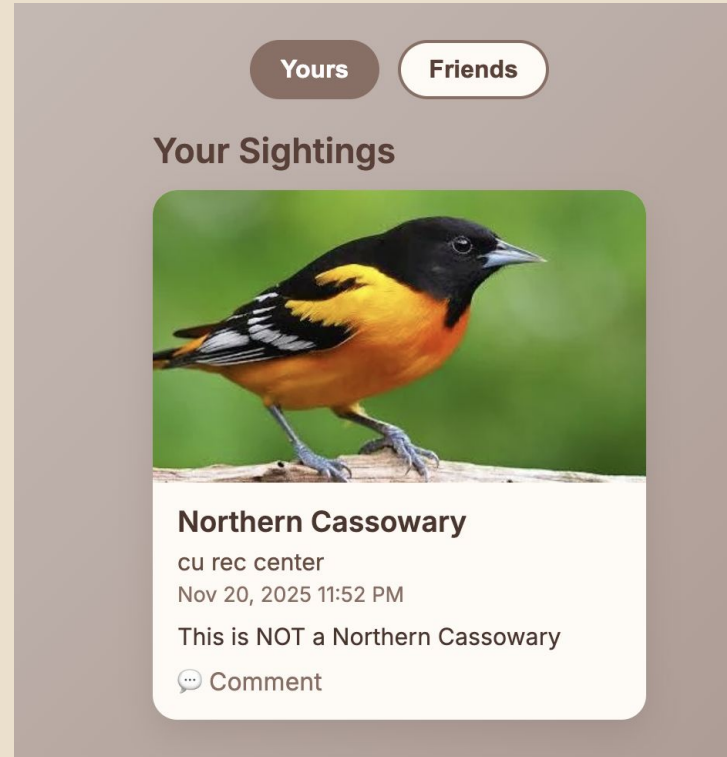
Sign Up

[Already have an account? Log in](#)

After signing in, users enter their own bird-tracking environment

The platform becomes a personal space for documenting sightings over time

A returning logbook rather than just a one-time interaction



Record bird sightings and attach notes or details

Store, revisit, and compare observations across time

Watch their personal list grow as more sightings are added

Fits both casual bird watchers and dedicated enthusiasts

Lightweight, organized, and built for ongoing exploration



Bird Name: Northern Cassowary

Location: cu rec center

Date: Thu Nov 20 2025 23:52:00 GMT+0000 (Coordinated Universal Time)

Description: This is NOT a Northern Cassowary

Posted by: Aaron Duong

Comment Here

Write your comment...

Post Comment

No comments yet. Be the first to comment!

Initial Planning:

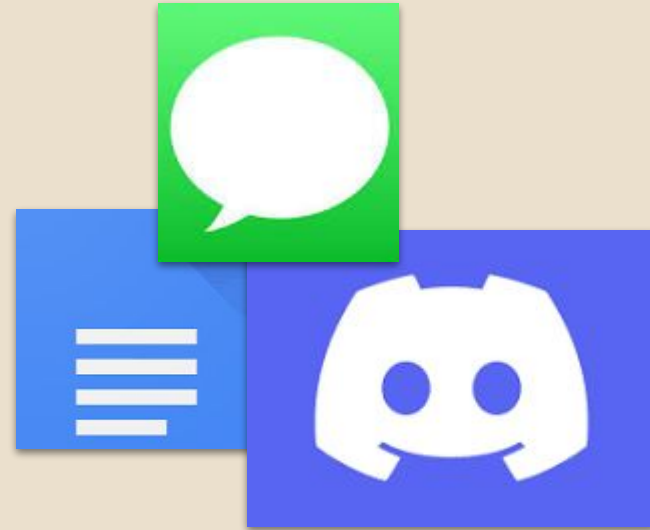
Brainstormed early concepts for app structure and needed features

Used Google Docs to outline scope, assign tasks, and record ideas

Used Discord to coordinate work sessions and progress updates

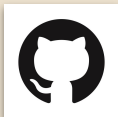
Accounted for scheduled meetings on Tuesdays and Thursdays

Helped keep track of tasks outside of scheduled project meetings



Tools Used

1



Project Tracker

Tool: GitHub Project Board

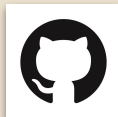
Purpose: Track user stories, tasks, weekly progress

Usefulness Rating:
★★★★☆ (4/5)

Methodologies Used:

- Iterative weekly planning
- Issue tracking + status tagging
- Task assignment per team member

2



VCS Repo.

Tool: GitHub

Purpose: Version control, branching, pull requests, merge flow

Usefulness Rating:
★★★★★ (5/5)

Methodologies Used:

- Feature branches per user story
- Pull requests + code reviews
- No direct commits to main

3



Database

Tool: PostgreSQL (Dockerized)

Purpose: Store user login data, hashed passwords, bird logs

Usefulness Rating:
★★★★☆ (4/5)

Methodologies Used:

- SQL schema planning early
- Init + insert scripts for deploy setup
- Persistent volume for local dev

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IDE

Tool: Visual Studio Code

Purpose: Main coding environment, UI, and Docker

Usefulness Rating:
★★★★★ (5/5)

Methodologies Used:

- Extensions for ESLint
- Live reload debugging
- Pair-programming when stuck

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UI Tools

Tools: Handlebars + HTML + CSS + JavaScript

Purpose: Render dynamic pages, layout structure, styling, interactions

Usefulness Rating:
★★★★☆ (4/5)

Methodologies Used:

- Incremental UI implementation
- Shared component partials

Tools Used

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App Server

Tool: Node.js

Purpose: Runs backend logic and app code

Usefulness Rating:
★★★★★ (5/5)

Methodologies Used:
- Modular routes
- Environment-based configuration

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Express JS

Framework

Tool: Express.js

Purpose: Routing, middleware, sessions, request/response control

Usefulness Rating:
★★★★☆ (4/5)

Methodologies Used:
- RESTful route structuring
- Middleware layering
- Session + auth pipeline

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Render

Deployment

Tool: Render

Purpose: Host + run app publicly for demo/presentation

Usefulness Rating:
★★★★☆ (3/5)

Methodologies Used:
- Deployment pipeline
- Environment variable configuration
- Testing after push

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API

eBird API

Purpose: Provides species data for classification

Mapbox API

Purpose: Displays bird sightings visually on a map

Gemini API

Purpose: Helps assist with bird analysis & suggestions

Usefulness Rating: ★★★★★☆ (4/5)

Methodologies Used:
- Implemented API calls using asynchronous requests, parsing returned JSON for integration into UI and logging features

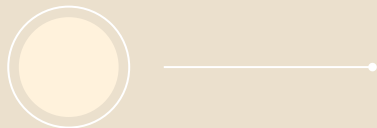
Testing (Lab 10)

Tools Used:

Mocha (Testing framework for structuring test suites & cases)

Chai / Chai-HTTP (Assertions + HTTP request testing)

Usefulness Rating: ★★★★★☆
(4/5)

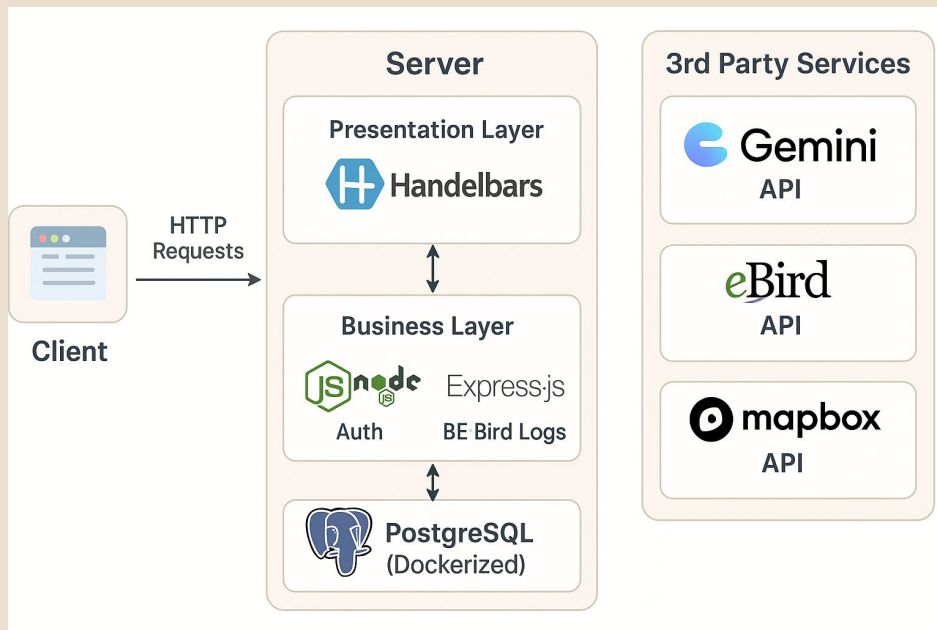


Purpose:

- Used to test the server backend and ensure routes respond correctly

- Validates HTTP endpoints, response codes, and returned data

Test Type	What It Validated
Default / Welcome Route	Ensured server responds 200 + success message
Register Route (Positive Case)	Successful user creation with valid data
Register Route (Negative Case)	Prevents empty/invalid registration requests



Architecture Diagram

Challenges Faced



Inconsistent team communication over time

Challenge 1: Communication and momentum slowed after the initial weeks

Challenge 2: Work happened at uneven times, making collaboration harder

How We Overcame It: Began checking GitHub commits more frequently, pulling updates often, and coordinating progress before pushing changes

Impact on Plans: Features took longer than expected and some tasks shifted to later sprints



Feature coordination and merging

Challenge 1: Branches conflicted or became out of sync during merges

Challenge 2: Explored extra tools/frameworks at one point, drifting from "simple first" planning

How We Overcame It: Reviewed changes before merging, tested locally, and focused development back on core functionality

Impact on Plans: Development pace slowed and scope realignment reduced available time for polish

Challenges Faced



API integration complexity (Gemini, Mapbox, eBird)

Challenge 1: API responses as well as values were unpredictable and needed reformatting

Challenge 2: Trial-and-error testing took time to stabilize requests and prompts

How We Overcame It: Adjusted request logic, refined parameters, and tested output formatting until consistent

Impact on Plans: Integration took longer than planned, pushing UI enhancements and extra features back

UI + functionality balance

Challenge 1: More UI ideas than time available

Challenge 2: Core features consumed most development focus early on

How We Overcame It: Prioritized core functions first (login, register, bird logging, map display) before UI styling

Impact on Plans: Final UI remains functional but visually simpler than originally envisioned

Enhancements (Additions)

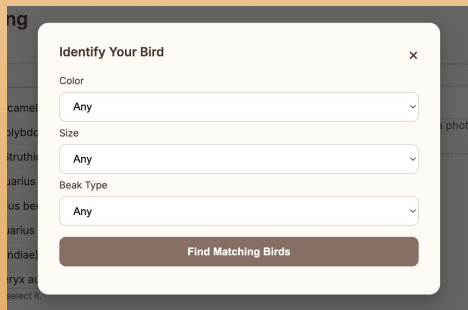
Advanced Gemini API Based Bird Identification Assistance

Expand prompt handling to accept more detailed user descriptions

Convert rough notes into species suggestions or likelihood ranges

Allow multi-input (color, location, behavior) to enhance prediction accuracy

Could evolve into a partial “field assistant” for quick referencing



A screenshot of a web form titled "Identify Your Bird" with a close button (X) in the top right corner. The form contains three dropdown menus labeled "Color", "Size", and "Beak Type", each with "Any" selected. Below the dropdowns is a brown button with the text "Find Matching Birds".

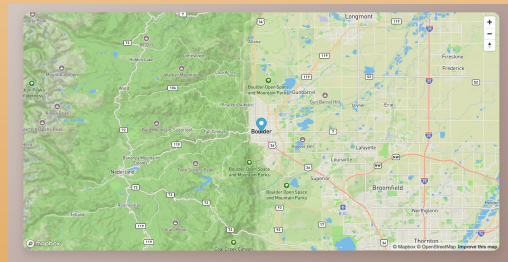
Map Filtering & Bird Visualization Tools

Filter the map by species, date, rarity, or region

Color-coded pins based on bird type or user frequency

Cluster or heat-map sightings for migration insights

Compare your sightings to community logs (if social features grow further)



Future Scope (New)



Aaron Duong



Northern Cassowary

Location: cu rec center

Date: 11/30/2025

This is NOT a Northern Cassowary

Comment

Improve Map Experience & Location Detail

Add support for manual pin adjustment or GPS-based auto-location

Allow sightings to store weather, time of day, or habitat context

Compare sightings over time to show movement or seasonal shifts



Aaron Duong

Email: aadu7536@colorado.edu

Member since: December 1, 2025

Bio: Not set yet.

My Posts



Northern Cassowary

This is NOT a Northern Cassowary

Location: cu rec center

Posted: Mon Dec 01 2025 03:52:20 GMT-

Comment

Personal Birding Statistics Dashboard

Track species count over time & generate user-level progress graphs

Display "first sightings," streaks, and rare observations

Provide badge-style achievements or goals to increase engagement

Demo: https://drive.google.com/file/d/1rOAa922UfnyuVtoYGObPd4R_10GYPgpS/view?usp=sharing



Thank You!